



Bayesian Statistics

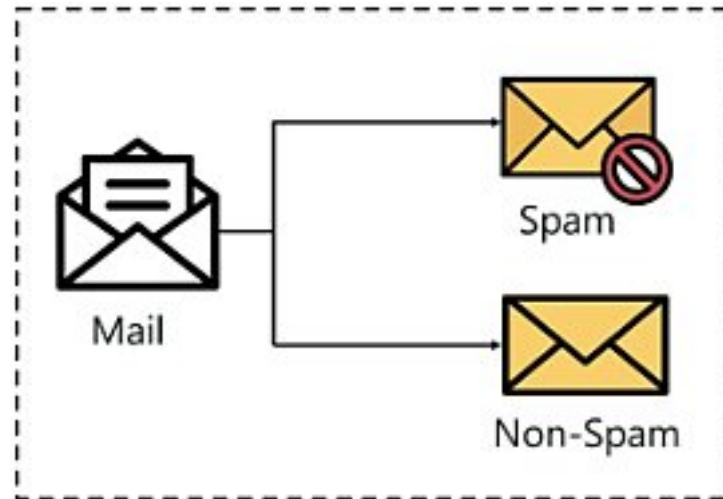


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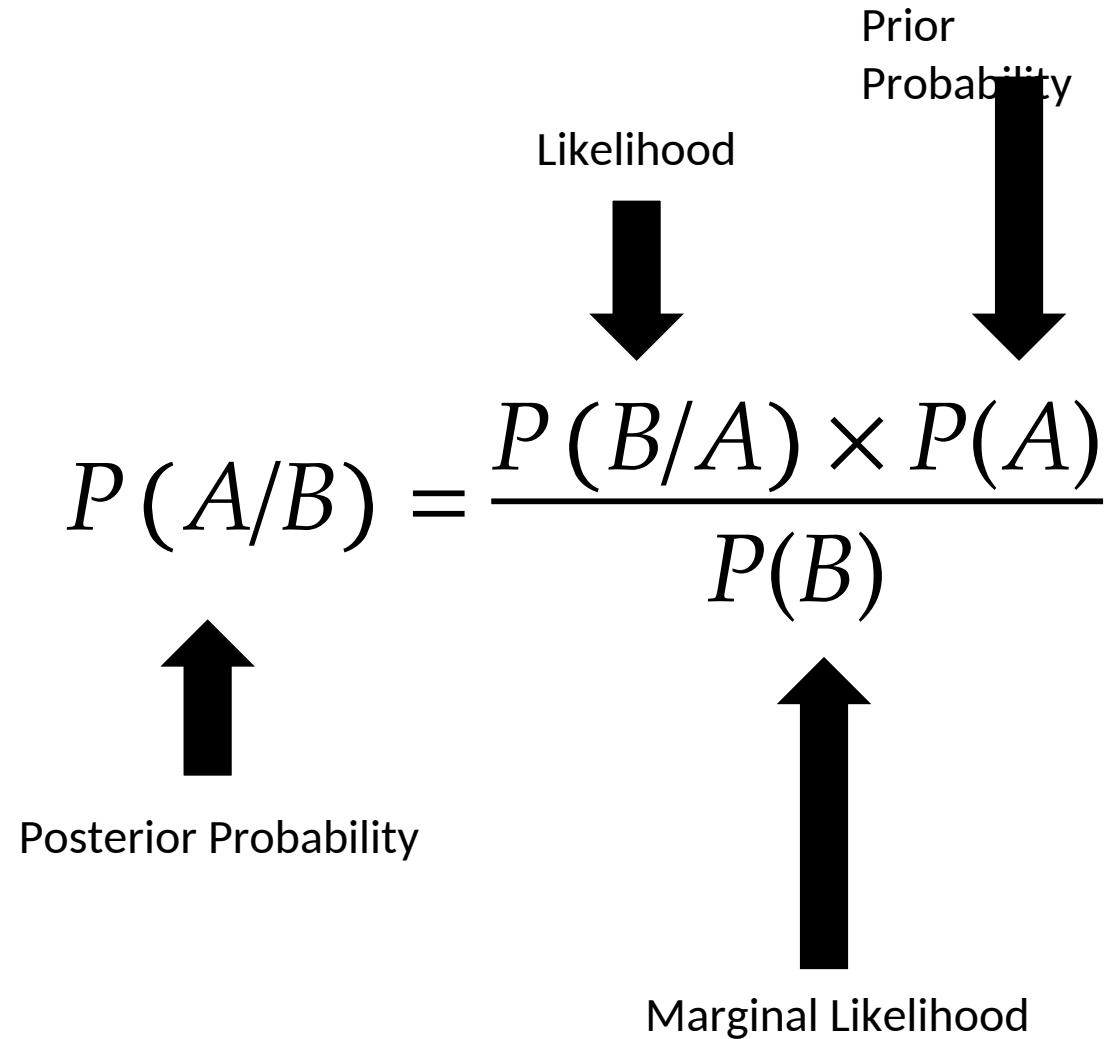
Classification

Classification in statistics or in data science refers to the process of categorizing data into predefined groups or classes based on certain criteria or characteristics.

Ex:-



Bayes' Theorem



The diagram illustrates Bayes' Theorem with the following components and arrows:

- Likelihood**: An arrow points down to $P(B/A)$ in the numerator.
- Prior Probability**: An arrow points down to $P(A)$ in the numerator.
- Posterior Probability**: An arrow points up to $P(A/B)$ on the left side of the equation.
- Marginal Likelihood**: An arrow points up to $P(B)$ in the denominator.

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

Sample Dataset

M1	Weather	Rank	Preparation	Injured or not	Win or not
1	Rain	bad	high	no	yes
2	snow	average	normal	yes	yes
3	clear	bad	light	no	no
4	clear	good	light	yes	yes
5	snow	good	normal	no	no
6	rain	average	light	no	no
7	rain	good	normal	no	no
8	snow	bad	high	no	yes
9	clear	good	high	yes	no
10	clear	bad	high	yes	yes

Application of Dataset

Now, with regards to our dataset, we can apply Bayes' theorem in following way:

$$P(Y/X) = \frac{P(X/Y) \times P(Y)}{P(X)}$$

Application of Dataset

- where, y is class variable and X is a dependent feature vector where:

$$X = (x_1, x_2, \dots, x_k)$$

- Just to clear, an example of a feature vector and corresponding class variable can be:

$$X = (Rain, Bad, High, No) \qquad Y = (Yes)$$

- $P(Y/X)$ here means, the probability of “having win” given that the weather conditions are “Rainy “, rank is “bad”, preparation level is “high” and “no injured players”.

Naive Assumption

- Independence among the features. So now, we split evidence into the independent parts.

$$P(A \text{ and } B) = P(A) * P(B)$$

- Hence, we reach to the result:

$$P(y|x_1, x_2, \dots, x_k) = \frac{P(x_1 | Y) * P(x_2 | Y) \dots P(x_k | Y) * p(Y)}{P(x_1) * P(x_2) * \dots * P(x_k)}$$

Naive Assumption

- Now, as the denominator remains constant for a given input, we can remove that term:

$$P(y|x_1, x_2, \dots, x_k) \propto p(x_1 | Y) * P(x_2 | Y) \dots P(x_k | Y) * p(Y)$$

- Create a classifier model.
- For this, find the probability of given set of inputs for all possible values of the class variable y and pick up the output with maximum probability.

$$y = \operatorname{argmax}_y \prod_{i=1}^k p(x_i | y) P(y)$$

- $P(y)$ is also called class probability and $P(x_i/y)$ is called conditional probability.

Prior Probability

- There are 10 data points.
- There are 5 instances of class/label 'yes', and 5 instances of class/label 'no'.
- The prior probabilities can be computed using the equation for prior probability

$$P(\text{Win} = \text{Yes}) = 5/10$$

$$P(\text{Win} = \text{No}) = 5/10$$

Class Conditional Probability - Weather



	Feature class count	Target class/label	Target class count
rain	2	No	5
snow	1	No	5
clear	2	no	5
Rain	1	Yes	5
Snow	2	Yes	5
clear	2	Yes	5

$$P(\text{Weather}_{\text{Rain}}/\text{Win}_{\text{No}}) = \frac{2}{5}$$

$$P(\text{Weather}_{\text{Snow}}/\text{Win}_{\text{No}}) = \frac{1}{5}$$

$$P(\text{Weather}_{\text{Clear}}/\text{Win}_{\text{No}}) = \frac{2}{5}$$

$$P(\text{Weather}_{\text{Rain}}/\text{Win}_{\text{Yes}}) = \frac{1}{5}$$

$$P(\text{Weather}_{\text{Snow}}/\text{Win}_{\text{Yes}}) = \frac{2}{5}$$

$$P(\text{Weather}_{\text{Clear}}/\text{Win}_{\text{Yes}}) = \frac{2}{5}$$

Class Conditional Probability - Rank



	Feature class count	Target class/label	Target class count
good	3	No	5
bad	1	No	5
average	1	no	5
good	1	Yes	5
bad	3	Yes	5
average	1	Yes	5

$$P(Rank_{Good}/Win_{No}) = \frac{3}{5}$$

$$P(Rank_{Bad}/Win_{No}) = \frac{1}{5}$$

$$P(Rank_{Average}/Win_{No}) = \frac{1}{5}$$

$$P(Rank_{Good}/Win_{Yes}) = \frac{1}{5}$$

$$P(Rank_{Bad}/Win_{Yes}) = \frac{3}{5}$$

$$P(Rank_{Average}/Win_{Yes}) = \frac{1}{5}$$

Class Conditional Probability - Preparation



	Feature class count	Target class/label	Target class count
high	1	No	5
normal	2	No	5
light	2	no	5
high	3	Yes	5
normal	1	Yes	5
light	1	Yes	5

$$P(\text{Preparation}_{High}/\text{Win}_{No}) = \frac{1}{5}$$

$$P(\text{Preparation}_{Normal}/\text{Win}_{No}) = \frac{2}{5}$$

$$P(\text{Preparation}_{Light}/\text{Win}_{No}) = \frac{2}{5}$$

$$P(\text{Preparation}_{High}/\text{Win}_{Yes}) = \frac{3}{5}$$

$$P(\text{Preparation}_{Normal}/\text{Win}_{Yes}) = \frac{1}{5}$$

$$P(\text{Preparation}_{Light}/\text{Win}_{Yes}) = \frac{1}{5}$$

Class Conditional Probability - Injured



	Feature class count	Target class/label	Target class count
yes	1	No	5
no	4	No	5
yes	3	Yes	5
no	2	Yes	5

$$P(\text{Injured}_{\text{Yes}}/\text{Win}_{\text{No}}) = \frac{1}{5}$$

$$P(\text{Injured}_{\text{No}}/\text{Win}_{\text{No}}) = \frac{4}{5}$$

$$P(\text{Injured}_{\text{Yes}}/\text{Win}_{\text{Yes}}) = \frac{3}{5}$$

$$P(\text{Injured}_{\text{No}}/\text{Win}_{\text{Yes}}) = \frac{2}{5}$$

Naive Bayes' Prediction

- Suppose we are now given a new feature vector:

- Weather condition: **rain**
- Rank level: **good**
- Preparation level: **normal**
- Presence of injured players: **no**



Lose (No win)

$$P(\text{Win}_{\text{Yes}}/\text{GivenFeatures}) =$$

$$P(\text{Weather}_{\text{Rain}}/\text{Win}_{\text{Yes}}) \times P(\text{Rank}_{\text{Good}}/\text{Win}_{\text{Yes}}) \times P(\text{Preparation}_{\text{Normal}}/\text{Win}_{\text{Yes}}) \times P(\text{Injured}_{\text{No}}/\text{Win}_{\text{Yes}}) \times P(\text{Win} = \text{Yes}) =$$

$$\frac{1}{5} \times \frac{1}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{10} = \frac{10}{6250}$$

$$P(\text{Win}_{\text{No}}/\text{GivenFeatures}) =$$

$$P(\text{Weather}_{\text{Rain}}/\text{Win}_{\text{No}}) \times P(\text{Rank}_{\text{Good}}/\text{Win}_{\text{No}}) \times P(\text{Preparation}_{\text{Normal}}/\text{Win}_{\text{No}}) \times P(\text{Injured}_{\text{No}}/\text{Win}_{\text{No}}) \times P(\text{Win} = \text{No}) =$$

$$\frac{2}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{5}{10} = \frac{240}{6250}$$

Naive Bayes' Prediction With Continuous Features

- Assume that the win prediction should be done based on the temperature in the ground area.
- Then,

$$P(\text{Temperature}/\text{Win}_{Yes}) = \frac{1}{\sigma_{\text{Temperature}/\text{Win}_{Yes}} \times \sqrt{2\pi}} \times e^{-\frac{1}{2} \times \left(\frac{\text{Temperature} - \mu_{\text{Temperature}/\text{Win}_{Yes}}}{\sigma_{\text{Temperature}/\text{Win}_{Yes}}} \right)^2}$$

$$P(\text{Temperature}/\text{Win}_{No}) = \frac{1}{\sigma_{\text{Temperature}/\text{Win}_{No}} \times \sqrt{2\pi}} \times e^{-\frac{1}{2} \times \left(\frac{\text{Temperature} - \mu_{\text{Temperature}/\text{Win}_{No}}}{\sigma_{\text{Temperature}/\text{Win}_{No}}} \right)^2}$$

- Then follow the process as previous.